DataTürk Project

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***Abstract*—This document is the first progress report for DataTürk's machine learning project. The document contains the details of the dataset cleansing process and the summary of the given dataset.**

***Index Terms*—machine learning, github, dataset, missing values, graphs**

I. Introduction

We have created a python script, which can be found on our repo [1], to identify the missing values in our given dataset, and analyze the data according to their contextual meaning.

II. Data Preprocessing

A. Dropping Some of the Columns and Filling the Values

We identified the columns that contained more than 50 percent of the missing data values in them and filled in these numeric columns using the calculated mean of such columns. For non-numeric columns we used the mode. In order to eliminate the bugs in the code, we took advantage of some sources and ChatGPT-4.

B. Creating a Binary Label for the Data

We converted the last columnt to a binary label column, which gave us a way to tag and identify these columns.

C. Summary of the Dataset

Total number of missing values in the DataFrame: 0
Total number of missing values in binary label column: 0
Non-numerical columns number is: 13
Numerical columns number is: 68
Total number of rows in the DataFrame: 246008
Total number of columns in the DataFrame: 81

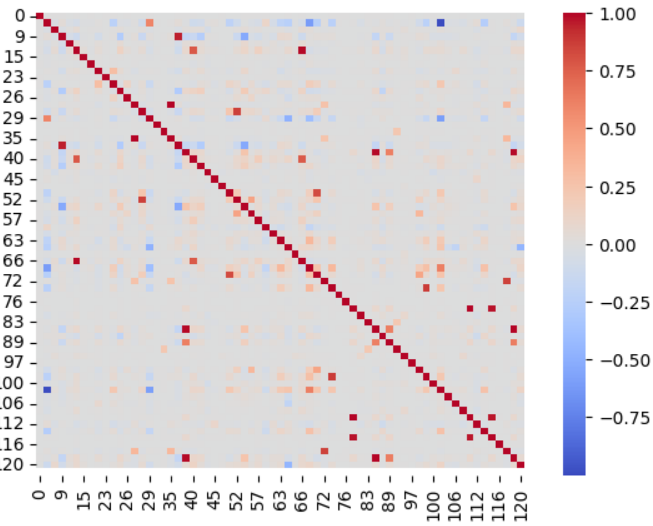

Fig. 1. Correlation between information on the dataset.

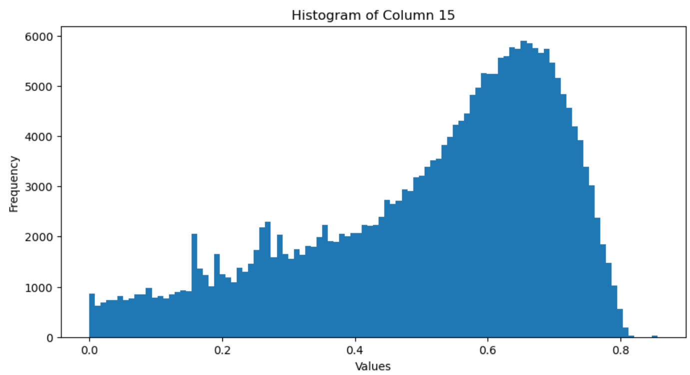

Fig. 2. Histogram of Column 15.

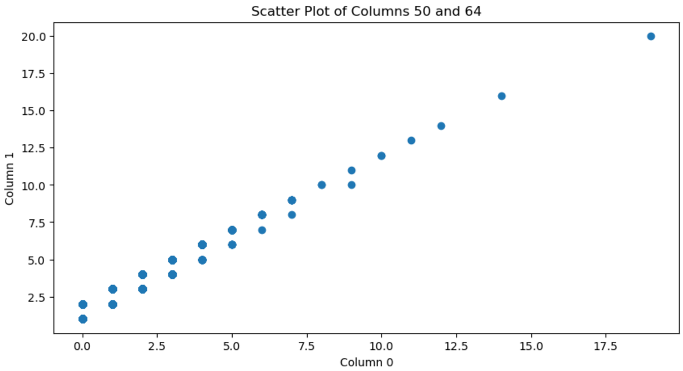

Fig. 3. Positive correlation between the column 50 and 64.

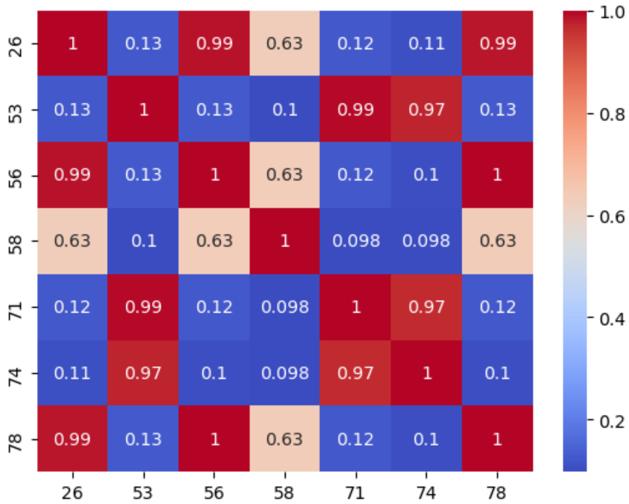

Fig. 4. .

REFERENCES

[1] iamemirsahin. DataTürk. [Online]. Available: https://github.com/iamsahinemir/DataTurk